

[This question paper contains 6 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 792 G
Unique Paper Code : 234403
Name of the Paper : Data Communication and Computer Networks (CSHT-409)
Name of the Course : B.Sc. (H) Computer Science
Semester : IV
Duration : 3 Hours Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
 2. **Question No. 1** is compulsory.
 3. Attempt any 4 of Question Nos. 2 to 7.
 4. Parts of a question must be answered together.
-
1. (a) What do you mean by a protocol ? What is the job of IP protocol in TCP/IP reference model ? (1+3)
 - (b) Assume a data stream is made of ten 0s. Encode this stream using the RZ encoding scheme. (2)

P.T.O.

(c) The following data pattern occurs in the middle of a data stream. A B ESC C ESC FLAG FLAG D

Give the output after byte stuffing. (2)

(d) A noiseless 4 kHz channel is sampled every 125 μ secs and every sample is encoded into 8 bits. What is the data rate? Also comment on the sampling rate. (3)

(e) What is the advantage of piggybacking scheme for handling acknowledgements in full duplex data communication. Is there any limitation of this scheme? How is it resolved? (4)

(f) Suppose instead of using 16-bit for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been? (2)

(g) What do you understand by CSMA/CD? (3)

(h) What is piggybacking? Give its advantage. (2+1)

(i) Discuss the constraints of Microwave Transmission. (2)

(j) Explain three way handshake process used in TCP. (4)

(k) What is the purpose of GET, PUT and POST methods of HTTP protocol? (3)

(l) **Explain** the working of DHCP protocol? (3)

2. (a) Generate even parity Hamming code for data bits 1101111. (4)

(b) List four differences between connection oriented and connection less services for data communication. (4)

(c) What is the use of Domain Name System? (2)

3. (a) Match the following to one or more layers of the OSI model (4)

(i) Transmission of bit stream across physical medium

(ii) Defines frames

(iii) Reliable process-to-process message delivery

(iv) Route Selection

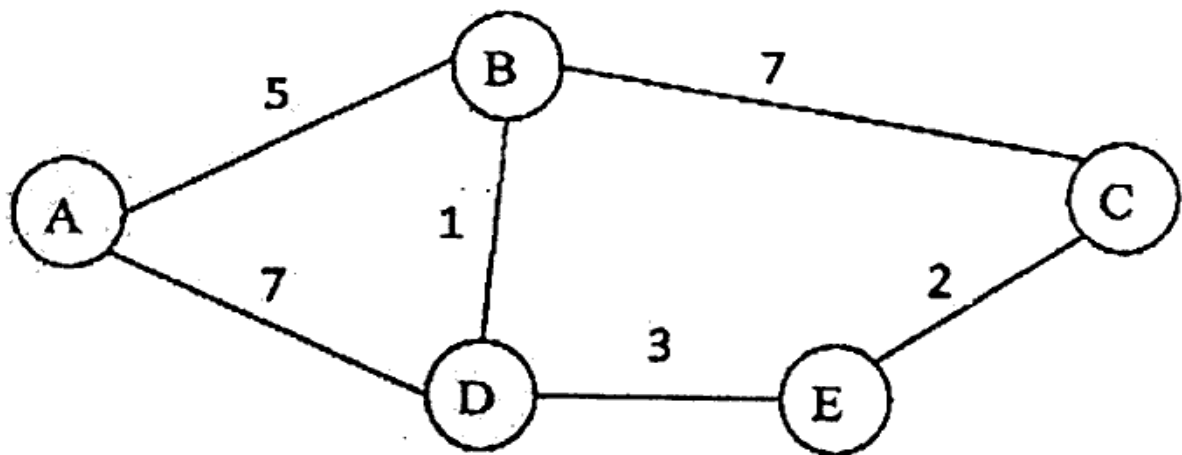
(b) Suppose you want to transmit the message 11001001 and protect it from errors using the CRC polynomial $x^3 + 1$. Determine the transmitted message. (4)

- (c) What is the baud rate of classic 10-Mbps Ethernet that uses Manchester encoding scheme? (2)
4. (a) Draw the graphs representing the following encoding schemes for the data 01010101 assuming that the last signal level has been positive. (4)
- (i) NRZ-L
- (ii) Differential Manchester
- (b) Minimum length of Ethernet frame is 64 bytes. Justify. (4)
- (c) Give the constellation diagram for QAM-16. (2)
5. (a) For selective repeat ARQ, what should be the maximum size of receiver's window, and why? (3)
- (b) Describe the problem of "Count to infinity" in Distance Vector Routing. Explain with a suitable example. (4)
- (c) Elaborate the process of releasing a connection in TCP. (3)

6. (a) **What is** the use of DF and MF flags in the IP header? (2)

(b) What is the use of URG and RST flag bits in TCP header? (2)

(c) Using Dijkstra's shortest path algorithm, find the shortest path from A to C. (4)



(d) Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link in there is a need for a guard band of 10-kHz between the channels to prevent interference? (2)

7. (a) What is an URL? Give an example to explain its parts. (4)

- (b) What are the advantages of Optical Fiber over other guided media? (4)
- (c) List any two problems of Remote Procedure Call. (2)

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